

End Semester Project

Current divider

Objective:

The objective of this project is to design, simulate, and analyze current divider circuits using parallel resistors and an LM358 operational amplifier. The project is developed to study how electric current divides between different branches according to resistance values. It also helps in understanding Ohm's Law, Kirchhoff's Current Law (KCL), comparator operation, and practical circuit implementation through both simulation and hardware testing.

Introduction:

A current divider is a basic electronic circuit in which electric current splits into multiple parallel branches. The amount of current flowing through each branch depends upon the resistance value of that branch. In this project, equal $10\text{k}\Omega$ resistors are used so that current divides almost equally between the branches. The circuit uses a 12V power supply and an LM358 operational amplifier as a comparator. LEDs are connected at the output to indicate the comparator state visually. The circuit was first designed and simulated in Proteus software and then implemented practically on a breadboard for verification.

Electronic Components:

1. Battery (12V)
2. LM358 / LM358N
3. Resistors ($10\text{k}\Omega$).
4. Potentiometer (RV2)
5. LEDs
6. $1\text{k}\Omega$ Resistors

Concept Used:

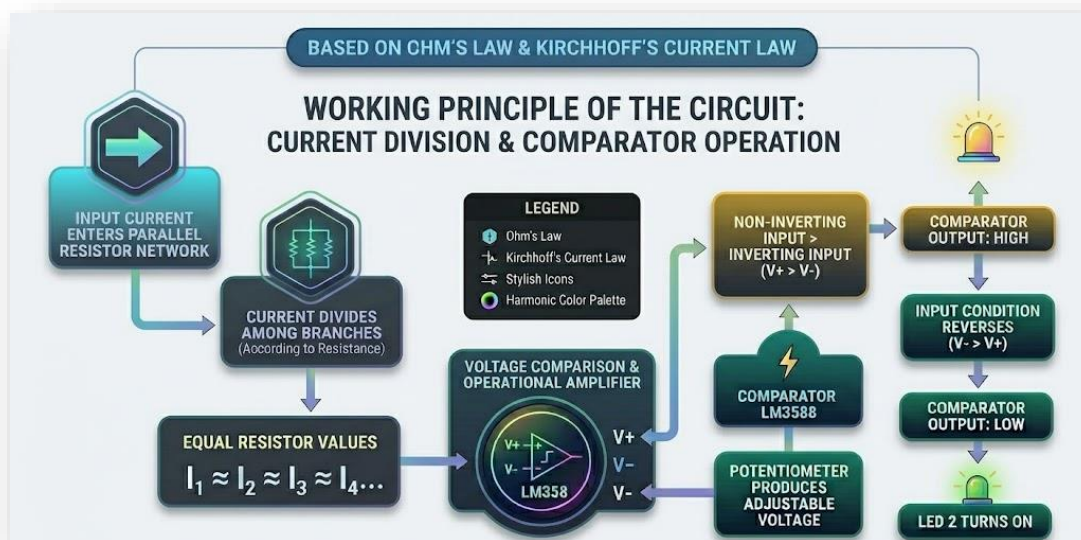
The project is based on the concept of current division in parallel resistor networks. Parallel resistors are used to divide the total current into different branches. A potentiometer is included to generate a variable reference voltage for the comparator. The LM358 operational amplifier compares the input voltages and changes its output state accordingly. LEDs are connected to the output to show HIGH and LOW conditions of the comparator.

Working Principle:

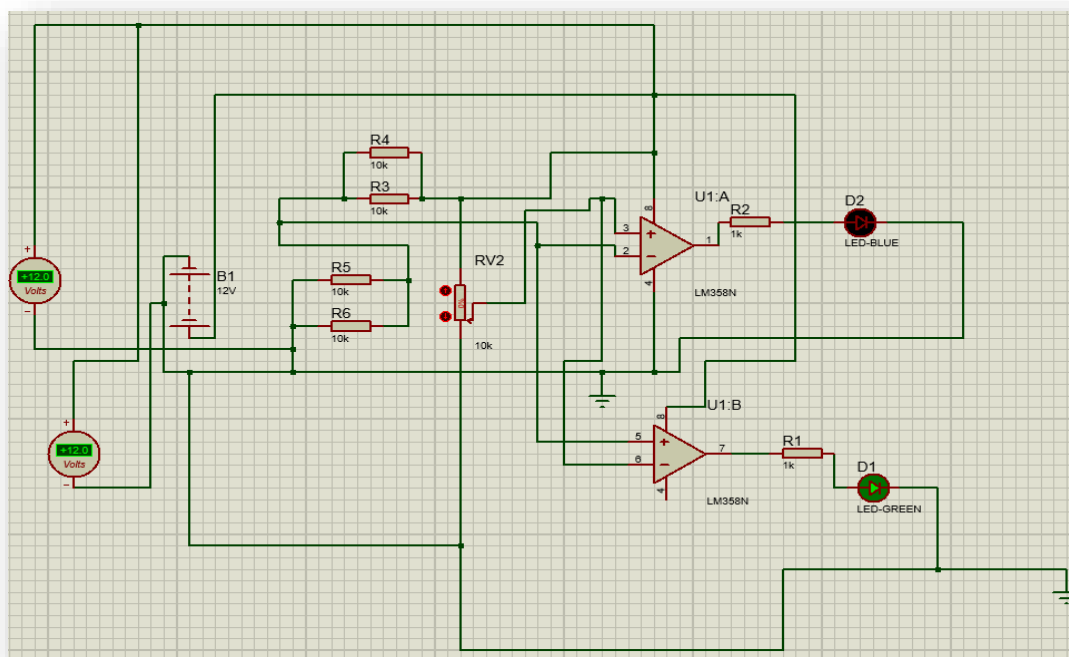
The working principle of this circuit is based on Ohm's Law and Kirchhoff's Current Law (KCL). When electric current from the 12V supply enters the parallel resistor network, it divides into multiple branches according to the resistance values connected in the circuit. Since equal $10\text{k}\Omega$ resistors are used in this project, the current is distributed almost equally between the branches, creating a balanced current divider network. The resistor arrangement also develops specific voltage levels at different nodes of the circuit. A potentiometer is connected to providing an adjustable reference voltage, allowing the user to manually vary the input signal and observe different output conditions.

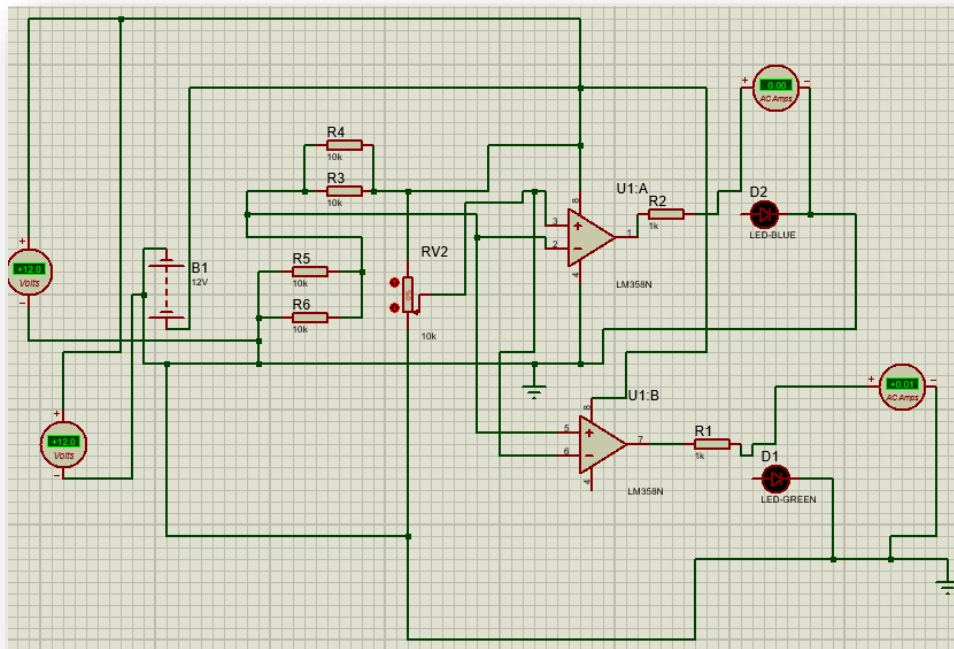
The LM358 operational amplifier is configured as a comparator that continuously compares the voltages applied at its non-inverting (+) and inverting (–) terminals. When the voltage at the non-inverting input becomes greater than the voltage at the inverting input, the output of the comparator switches to a HIGH state close to the supply voltage, causing one LED to glow while the other remains OFF. Similarly, when the voltage at the inverting input becomes greater, the comparator output changes to a LOW state, turning OFF the first LED and turning ON the second LED. This switching action clearly demonstrates the operation of comparator circuits along with the practical behavior of current division in parallel resistor networks. The circuit was verified through both Proteus simulation and practical breadboard implementation, where the observed results matched the theoretical calculations.

Flow chart:



Simulation:





Current is divided as upper led is off and bottom one is on

Advantages:

This circuit is simple, low cost, and easy to implement. It provides a clear understanding of current division and comparator operation. The project is suitable for beginners in electronics and can easily be simulated and tested practically. It also helps in understanding resistor networks, current flow, and operational amplifier applications.

Disadvantages:

The circuit performance depends on accurate resistor values, and small variations in resistance can affect current distribution. It is not suitable for high-power applications and may be sensitive to component tolerances and loose connections during practical implementation.

Conclusion:

This project successfully demonstrates the principle of current division using parallel resistors and comparator operation using the LM358 operational amplifier. The Proteus simulation and practical breadboard implementation verified the theoretical behavior of the circuit. The resistor network divided the current effectively, while the comparator-controlled LED switching according to voltage conditions. The project improved understanding of current division, Ohm's Law, Kirchhoff's Current Law, comparator circuits, and practical electronics implementation.

References:

1. [Proteus 8.6 User Guide and Simulation Software.](#)
2. [Texas Instruments, LM358 Datasheet.](#)
3. [Electronic Devices and Circuit Theory.](#)
4. [Principles of Electric Circuits.](#)
5. [Basic concepts of Ohm's Law and Kirchhoff's Current Law \(KCL\).](#)